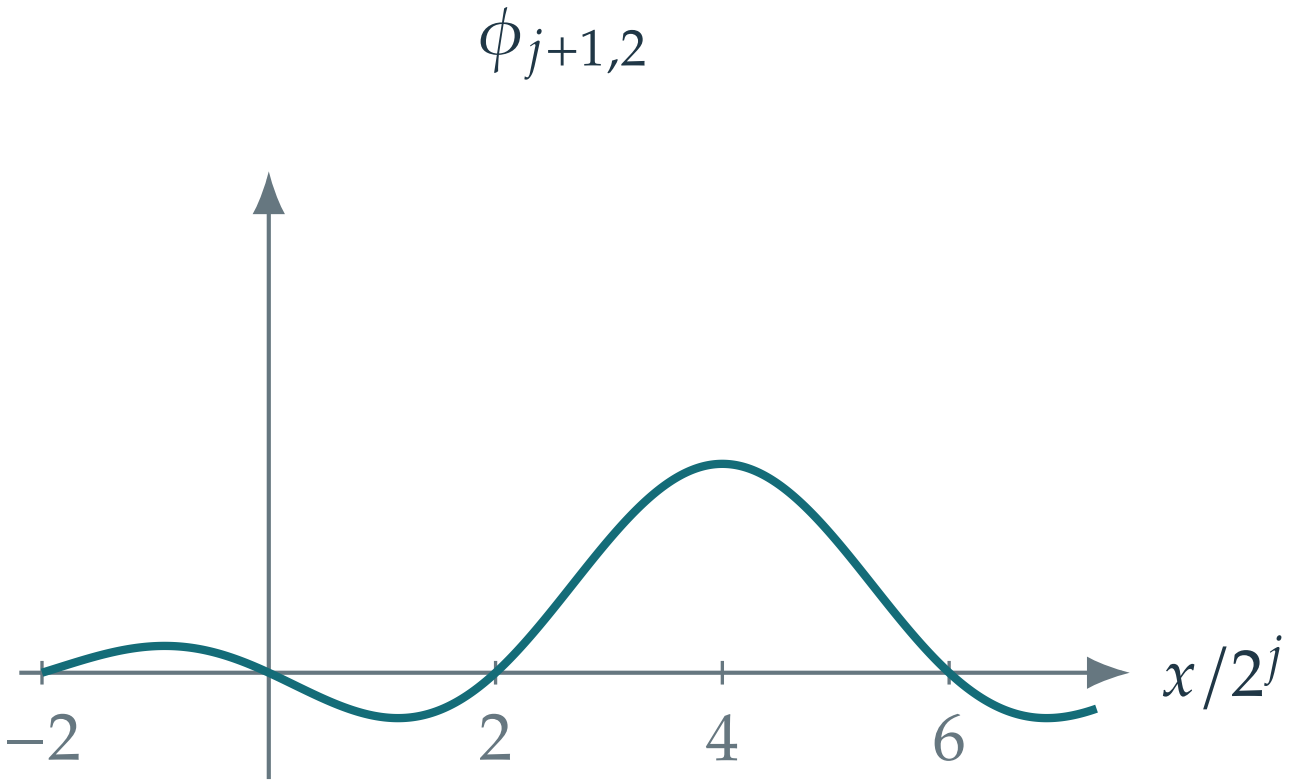

center $4 \cdot 2^j$, scale 2^{j+1}
relative peak $1/\sqrt{2}$